

Peripheral Driver with Fixed Regulators 78xx and 79xx

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Drivers for peripheral devices should have over-current, over-voltage and over-heating protections. Nowadays, a large variety of low-cost, fixed-voltage regulators (such as 78xx and 79xx series) are available which offer these protections and more. These can be used as electronic on/off switches and drivers for such peripheral devices as DC motors, incandescent lamps, electromagnet relays and power LEDs, besides others. The drivers can be with single polarity, dual polarity, half bridge and full bridge. This article discusses one such application.

Circuit and working

Fig. 1 shows circuit diagram of a bipo-

lar half-bridge driver built using ICs 7812 and 7912, but it can also be built with other similar regulators. The circuit will work with any 78xx and 79xx series of regulators using 5V to 24V supply. The circuit works with symmetrical as well as non-symmetrical voltages. For example, we can have $\pm 12V$, or $+12V$ and $-9V$ (or any other voltages within the regulators' range), if needed.

In this case, IC 7812 (IC1) works as a switch and regulator for the positive input voltage V_1 , while IC 7912 (IC2) works as switch for the negative input voltage V_4 . Diodes D1 through D4 protect the regulators from reverse voltages. Diodes D3 and D4 are obligatory only if there is a danger of reverse voltage from the capacitors connected to the regulators.

The $+V_2$ voltage is controlled by transistors T1 and T2. When T1 and T2 are on, voltages V_2 , V_3 and V_7 are present. In that case, you have $V_3 = 12V$ and $V_7 = 12V - 0.7V = 11.3V$. Similarly, when T1 and T2 are off, voltages V_2 , V_3 and V_7 become zero. When the signal at point A on input connector CON1 is high positive (say, above 3V), transistors T1 and T2 are on. When the signal at point A is low or negative (say, below 0.8V), transistors T1 and T2 are off.

The $-V_5$ voltage is controlled by transistors T3 and T4. When T3 and T4 are on, the voltages $-V_5$, $-V_6$ and V_7 are present. In that case you have $-V_6 = -12V$ and $V_7 = -12V + 0.7V = -11.3V$. When T3 and T4 are off, the voltages

$-V_5$, $-V_6$ and V_7 are zero. When the signal at point B is highly negative (say, $-3V$ or less) the transistors T3 and T4 are on. When the signal at point B is not so negative (say, $-0.7V$ or more), transistors T3 and T4 are off.

If you activate points A and B at the same time, voltage V_7 will be zero. The control signal at point A can be made TTL or CMOS compatible with selection of appropriate resistors R1 and R2 and transistor T1. The control signal at point B is negative and is not TTL or CMOS compatible.

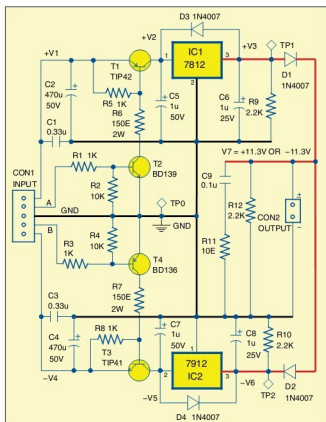


Fig. 1: Circuit diagram of peripheral driver with fixed regulators

Test Points

Test point	Details
TP0	0V
TP1	+12V
TP2	-12V

PARTS LIST

Semiconductors:

- IC1 - 7812, 5V regulator
- IC2 - 7912, -12V regulator
- T1 - TIP42 pnp transistor
- T2 - BD139 npn transistor
- T3 - TIP41 npn transistor
- T4 - BD136 pnp transistor
- D1-D4 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1, R3 - 1-kilo-ohm
- R2, R4 - 1-kilo-ohm
- R5, R8 - 1-kilo-ohm
- R6, R7 - 150-ohm, 2W
- R9, R10, R12 - 2.2-kilo-ohm
- R11 - 10-ohm

Capacitors:

- C1, C3 - 0.33uF ceramic disk
- C2, C4 - 470uF, 50V electrolytic
- C5, C7 - 1uF, 50V electrolytic
- C6, C8 - 1uF, 25V electrolytic
- C9 - 0.1uF ceramic disk

Miscellaneous:

- CON1 - 5-pin connector
- CON2 - 2-pin connector